



Case Study: Query Practice

SAP Business One 10.0, version for SAP HANA

PUBLIC

INTRODUCTION

These practice exercises are designed to give you hands-on use in creating basic SQL queries for SAP HANA using the built-in query tools in SAP Business One.

You will create the following queries:

1. A report showing a list of customers
2. A report that uses a parameter as selection criteria
3. A report based on multiple tables
4. A report with a running total that can be used with an alert to provide a worklist to a user
5. A query used in a dashboard widget

PREREQUISITE:

1. Use the demo database for SAP Business One 10.0, version for SAP HANA
2. Credentials: User code: **manager**

Use either of the SAP Business One query tools: **Query Generator** or **Query Wizard**. Note: the solutions are shown for the **Query Generator** only. All queries shown here use the HANA SQL syntax.

Important note: The reports included in this case study show data from the UK localization database. The data you will see will obviously be different depending on your localization and the date you run the queries.

Task 1

Create a Customer List Report

This report displays a list of customers and balances from the **OCRD** Business Partners table.

1. Create a query using the following fields from the OCRD Business Partners table: CardCode, CardName, Address, City (Bill-to city), ZipCode, Balance, CntctPrsn.
2. Since the OCRD table also contains records for vendors and leads you need to filter customer records using the CardType = 'C' condition.
3. In the query results, sort the results alphabetically by customer name and add a total to the account balances column.
4. Save the query in a new category called *Sales* and choose the *Assign Group* button to assign the category to the *Saved Queries – Group No.1* authorization group.
5. Run the saved query from the *Tools* menu.

Note: When using a HANA database, multi-case field names must be enclosed in double quote marks in the query.

The query and the results should look something like this:

Query Preview

```
SELECT T0."CardCode", T0."CardName", T0."Address", T0."Balance", T0."ZipCode", T0."City", T0."CntctPrsn"
FROM OCRD T0 WHERE T0."CardType" = 'C' ORDER BY T0."Balance"
```

#	BP Code	BP Name	Bill-to Street	Account Balance	Bill-to Zip Code	Bill-to City	Contact Person
1	C60000	SG Electronics	45th Street	-27,077.56	10010	New York	Eric Alexander
2	C23900	Parameter Technology	908 Darby Road	-15,356.78	B1 6TR	Birmingham	Daniel Brown
3	C40000	Earthshaker Corporation	94 Bolton Road	-1,653.03	NW12 8HG	London	Bob McKensly
4	C50000	ADA Technologies	FluggeStrasse 6	0.00	22303	Hamburg	Mary Brown
5	C1234	VIP		0.00			
6	C42000	Mashina Corporation	400 Range Road	36.00	M3 7FD	Glasgow	Anthony Smith
7	C99999	One Time Customer		463.09			
8	C25000	Star Company	Plynarenska	975.18	821 09	Bratislava	Martina Basalykova
9	C30000	Microchips	45th Street	1,392.00	N1 8GH	Leeds	Judy Brown
10	C99998	Web Customer		1,596.00			
11	C70000	Aquent Systems	Hollywood Ave	2,325.72	95112	San Jose	Troy Brown
12	C26000	River Ltd.	United Rd	5,094.00	M16 0RA	Manchester	Lewis Beerman
13	C20799	34 Technologies		13,200.00			
14	C20000	Maxi-Teq	Hendon Way	13,866.20	NW2 7YT	London	Norm Thompson
15	C80000	Pivotal Solutions	8th Avenue	50,217.77		New York	
				45,078.59			

Execute Cancel Reverse Table Copy Data Save Save As Open

Fine Tune the Results

- You can sort the results according to any column by **double-clicking** in the header field of the column.
- To include a total of the account balances, press **Ctrl and click** in the header field of the Account Balance column. The total will appear at the bottom of the column, as shown below.

Task 2

Create a Report with a Parameter

This report displays a list of customer invoices posted after a certain date. The date is entered as a parameter when the query is run. The report uses the OINV invoice table.

Find Field Names

To find out the field names for the report, use system information:

Name in Document	Database Field Name
<i>No.</i>	
<i>Customer</i>	
<i>Name</i>	
<i>Posting Date</i>	
<i>Total</i>	DocTotal

Note: When you hold your mouse over the Total field, the field name does not display in system information. This is because this field holds the currency symbol as well as the amount. The database field name is *DocTotal*.

Create the Query

Select the fields from the **OINV** table that you identified using system information.

Filter by open sales invoices and use the variable [%0] with the *Greater than* condition to filter only invoices posted later than a date entered by the user as a parameter:

T0."DocStatus" = 'O' and T0."DocDate" > [%0]

Run the Query

Choose *Execute*. Select a date from the list. **Note:** if you are using the demo database the invoices you see may be very old, unless you have recently added invoices.

To display the total for the invoices, choose Ctrl and double-click the *Document Total* column heading.

6	1308	➔ C80000	Pivotal Solutions	02.04.	50,217.77
4	1306	➔ C20799	34 Technologies	22.03.	13,200.00
3	1293	➔ C26000	River Ltd.	26.08.	3,402.00
5	1307	➔ C40000	Earthshaker Corporation	29.03.	2,700.00
7	1313	➔ C20000	Maxi-Teq	14.04.	2,000.00
1	1243	➔ C70000	Aquent Systems	26.08.	1,858.50
2	1254	➔ C40000	Earthshaker Corporation	26.08.	112.50

Save this query with the name **Invoice List** in the *Sales* category.

Task 3

Create a Report from Multiple Tables

This query will display a list of open sales quotations summarized by customer and grouped by sales employee. The query will use two tables:

- Sales Quotation (OQUT)
- Sales Employee table (OSLP)

The inner join will be provided for you by the query tool.

Create the query

Select the OSLP table and then select the **SlpName** field.

Select the OQUT table and then select the **CardCode** and **CardName** fields.

Calculate the total value of the sales quotations using the SUM function and provide a column heading in the report:

SUM(T0."DocTotal") as "Total Value"

Count the number of sales quotations for each customer using the COUNT function and provide a heading:

COUNT(T0. "DocNum") as "No of Documents"

Filter the query so that only open quotations are used:

T0."DocStatus" = 'O'

Sort by the **SlpName** field.

Group by **SlpName**, **CardCode** and **CardName** fields.

Run and save the query.

Task 4

Create a Report as a Worklist for a User

This report displays all the sales orders posted for today, organized by sales employee name. The report uses the ORDR table and the OSLP table. The report is saved then can be later used with an alert to provide a daily worklist for a user.

Preparation: Create 2-3 sales orders for customers, with today's posting date. Select a discount % in each order.

Create the query

Select the ORDR and OSLP tables.

Select the sales employee name, document number, card code, document total and discount. SUM the document total from each sales order, combined with the OVER clause to keep a running total for each sales employee.

Only select sales orders for the current day:

The screenshot shows the 'Query Generator' window with the following configuration:

- Tables:** ORDR and OSLP are selected in the left pane.
- Select:**

```
T1."SlpName", T0."DocNum", T0."CardCode",
T0."DocTotal", SUM(T0."DocTotal") over (Order by
T1."SlpName", T0."DocTotal") as "Running Total",
T0."DiscPrct"
```
- From:**

```
ORDR T0 INNER JOIN OSLP T1 ON T0."SlpCode" =
T1."SlpCode"
```
- Where:**

```
T0."DocDate" = Current_Date
```
- Sort By:** (Empty)
- Group By:** (Empty)
- Buttons:** Execute, Close, and Conditions.

Execute the query

Save the query in the *Sales* category with the name **Today's Sales Orders**.

INFORMATION: The report can be scheduled to run daily and sent to a user using the alert mechanism. This will be covered in the case study for alerts.

Task 5

Create a Query for a Dashboard Widget

Set up a count widget that counts the number of deliveries created each day.

Create the query

The query should select document numbers from the ODLN table where the document date (posting date) is the current date.

Save and name the query **Deliveries_Today**.

Test the Query

Create a few deliveries (with the current date) so that you can test your query.

Set up the Count Widget

Tools > Cockpit → Count Widget Setup

Switch to *Add* mode.

Give the count widget a code, name and description. The name will appear on the widget so make it short and precise. For example, name it **Deliveries Today**.

Link the count widget to the query you just created.

Choose *Add*.

Add the Count Widget to your cockpit

Choose the *Pencil* icon to make changes to your cockpit.



Find the newly created widget in the *Widget Gallery* (choose *Business Object Count* in the dropdown box) and add it to the cockpit.

Save the changes to the cockpit.



Test the Count Widget

You can test the count widget by adding more deliveries. You can wait until the cockpit picks up the new deliveries or chose *Refresh* to see the count updated.





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